

# CLOUD COMPUTING FUNDAMENTALS

## Course Description

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School of Engineering  
Universidad Distrital Francisco José de Caldas

2026-III



UNIVERSIDAD DISTRITAL  
FRANCISCO JOSÉ DE CALDAS

# Outline

- 1 You don't know who I am
- 2 Course Overview
- 3 Grading & Rules
- 4 Bibliography



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# Academic Experience

- **Computer Engineer**, M.Sc. in Computer Engineering, and *researcher* for **17 years**.
- 9 years as **full-time associate professor** at colleges, in **Computer Engineering programs**.
- 3 years as **lecturer professor** for both colleges and **government STEM programs**.
- Speaker at **IEEE** events and colleges in Colombia, Brazil, and Bolivia.



# Non-academic Experience



- **PyCon Colombia** and **Python Bogotá** co-organizer.
- 3 years as **software engineer** for several **tech companies** in Colombia.
- 3 years as **Technical Leader** of **Machine Learning and Data Science** at a USA startup.
- 1.5 years as **MLOps Engineer** for a **LATAM Fintech** company.
- Currently, **Senior Engineering Manager** of **Data Engineering** and **Machine Learning** at Blend 360.



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# Overview

This course is designed to **introduce undergraduate students** to **cloud architectures and administration** as part of the foundation for becoming an experienced *cloud engineer* or *DevOps professional* capable of designing *scalable IT solutions*.

The course starts with a **comprehensive introduction to cloud computing fundamentals and service models**. Then, it transitions into **cloud strategies, logging, and security design**. Finally, we will focus on **continuous evaluation, disaster recovery planning, and practical implementations** adopted by **Google, Amazon, and Microsoft**.

AWS Lambda → 10<sup>6</sup> Free/month

Classes will consist of **lectures, architecture design sessions, and cloud deployment projects**. Also, you must complete some readings from *cloud computing best practices* and *infrastructure as code*. In addition, there will be a **course-long project, two course tests, and practical workshops**.



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# Course Objectives

## General Objective

Train students in the design and administration of cloud solutions, enabling them to adapt to modern technological needs and contribute to the digital transformation of organizations through the efficient use of cloud infrastructures and services.

## Specific Objectives

- Identify and analyze the fundamental concepts, service models, and architecture of cloud computing.
- Apply industry-leading tools and platforms for the implementation of cloud solutions.
- Develop systems for the efficient management of data and applications in cloud environments, ensuring compliance with regulations and requirements.

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# Prerequisites

This is an advanced course, so you must have some foundational knowledge in:

- **Computer Networks** limits and design (IPs, DNS, HTTP, TCP/UDP).
- Operating System **App.** administration (Linux, permissions, processes).
- **Software Architecture** (APIs, web services).
- **Git** basic usage and **GitHub** fundamentals.
- Foundational understanding of **containers** (Docker) is highly desirable.

— router  
— switch  
Phys.



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30% ID

server → NO GUI  
console - linux



# Syllabus

| Period     | Topic                                  | Time        |
|------------|----------------------------------------|-------------|
| Period I   | Introduction to Cloud Computing        | 4 sessions  |
|            | Cloud Service Models                   | 5 sessions  |
|            | General Considerations & Auditing      | 12 sessions |
|            | — Mid-Term Test                        | 1 session   |
| Period II  | Cloud Strategy and Security            | 12 sessions |
|            | DevOps Culture & Continuous Evaluation | 12 sessions |
|            | — Final Test                           | 1 session   |
| Period III | — Application & Impact Evaluation      | 3 sessions  |



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# Grades Percentages

| Period     | Item                    | Percentage |
|------------|-------------------------|------------|
| Period I   | Workshops               | 15%        |
|            | Mid-Term Test           | 20%        |
| Period II  | Workshops               | 15%        |
|            | Final Test              | 20%        |
| Period III | Paper + Poster          | 5%         |
|            | Report + Implementation | 15%        |
|            | Presentation            | 10%        |

30%  
30%  
30%



# Don't hate the player, hate the game

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- Class attendance is **not mandatory**. If you miss classes, you must study *independently*.
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# Code of Conduct

- Always be **respectful** to your **classmates** and to me. You must be **kind** to everyone inside (*and outside*) the classroom.
- There is **no** best **programming language**, **tool**, or **technology**. There are only **better** or **worse** solutions.
- You must be **honest** with your work. If you **don't know something**, just **ask** me. I will be **glad** to help you.
- You must be **responsible** with your work. If you don't submit **on time**, please **don't complain**.
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Recommended bibliography:

- **Cloud Computing: Concepts, Technology & Architecture**, by Thomas Erl, Zaigham Mahmood, & Ricardo Puttini.
- **Designing Data-Intensive Applications**, by Martin Kleppmann.
- **The Phoenix Project**, by Gene Kim, Kevin Behr, & George Spafford.
- **Accelerate: The Science of Lean Software and DevOps**, by Nicole Forsgren, Jez Humble, & Gene Kim.
- **Site Reliability Engineering: How Google Runs Production Systems**, by Niall Richard Murphy, Betsy Beyer, Chris Jones, & Jennifer Petoff.



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# Thanks!

## Questions?



**My Profile:** [www.linkedin.com/in/casierrav](https://www.linkedin.com/in/casierrav)

